

Reinforcement Learning-Based Optimized Adaptive Secure Control for Constrained Fractional-Order Nonlinear Systems Under FDI Attacks

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Abstract—This article considers the adaptive fuzzy optimized secure self-triggered control (STC) problem for constrained fractional-order nonlinear systems (FONSSs) subject to unknown false data injection attacks (FDIAs). To fulfill the unilateral full-state constraints (UFSCs), an emerging nonlinear state-dependent function with convexity is utilized by means of its property in removing the feasibility conditions existing in the traditional constraint control. Meanwhile, since the true information of the state variables under attack signals is unavailable, a method of coordinate transformation combining the compromised system and dynamic surface control is used for designing the corresponding controller while alleviating the adverse impacts of the FDIAs. Additionally, by constructing the equivalent auxiliary systems and employing actor-critic neural networks (NNs), a reinforcement learning (RL)-based adaptive fuzzy optimal program is provided to achieve optimal control. Furthermore, considering that event-triggered mechanism needs real-time supervision of the control signals, a STC strategy is introduced to circumvent this drawback and reduce communication consumption. Leveraging the fractional Lyapunov stability theory, it has been confirmed that the devised controller can ensure that the stabilization errors tend toward a small area nearby the origin at the minimum costs and all the signals arising in the closed-loop system (CLS) are bounded. Eventually, two simulation examples are offered to demonstrate the reported control algorithm's validity.

Index Terms—Adaptive optimized control, false data injection attacks (FDIAs), fractional-order nonlinear systems (FONSSs), interval type-2 fuzzy logic systems (IT2FLSs), reinforcement learning (RL), unilateral full-state constraints (UFSCs).

I. INTRODUCTION

FRACTIONAL calculus, as an extended form of integer-order calculus, has been widely applied to model complex nonlinear systems in recent years owing to its special historical memory properties [1], [2]. Therefore, the control issues about fractional-order nonlinear systems (FONSSs) have been favored among scholars and tremendous research results have been published, such as backstepping control [3], fuzzy control [4], and sliding mode control [5]. Wherein, adaptive backstepping control is an efficacious means by taking the intermediate variables as virtual control signals [6]. However, it has been widely recognized that the potential computational complexity problem generated by the iterative differentiation for the virtual controller will hinder the application of this recursive control technique. To solve this trouble, dynamic surface control solutions were presented via incorporating a nonlinear filter into the recursive design framework [7], [8]. Undeniably, the nonlinear dynamic is ubiquitous in practical applications. Accordingly, intelligent control methods involving fuzzy logic systems and neural networks (NNs) have been employed to model unknown nonlinear terms [9], [10], [11]. To be specific, [10] presented the adaptive fuzzy backstepping control tactic for FONSSs, where interval type-2 fuzzy logic systems (IT2FLSs) were adopted to tackle the uncertainty. Hereafter, an adaptive NN backstepping control solution was reported for FONSSs in [11]. Yet, the question about how to achieve a balance between system performance and resource consumption has not been fully considered therein.

As one of the current mainstream research directions, optimal control is achieved by minimizing the performance index, which centers on solving the Hamilton–Jacobi–Bellman (HJB) equation. Nevertheless, it is difficult to find an analytical solution directly because of the nonlinear features of HJB. Fortunately, reinforcement learning (RL) scheme has been proven to be an efficient tool to tackle optimal control issues, which not only can derive an approximate solution via iterative approximation online, but also can eradicate the

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dimensionality catastrophe inherent in dynamic programming approach [12], [13]. So far, many research works on RL strategies have been produced [14], [15], [16]. In detail, an RL-based fuzzy adaptive optimal control scheme was provided about high-order dynamic systems in [15]. Thereafter, [16] employed the actor–critic architecture to handle the robust stabilization problem related to nonlinear systems. Nonetheless, due to the complex form of FO calculus, it is difficult for FONSSs to directly achieve the optimal control by these classical methods. To this end, a fractional Euler–Lagrange equation was achieved via utilizing variational method in [17]. Nevertheless, solving the above equation is quite tricky and the system also needs to be fully known. For this problem, by establishing an auxiliary system and equivalent change, a feasible RL scheme was offered concerning FONSSs in [18]. Noteworthy, the optimized adaptive secure control issues for FONSSs under cyberattack has not been fully investigated in the existing relevant literature so far.

In modern engineering systems, the openness of network communication channels makes the control systems vulnerable to various types of network attacks, such as replay attacks [19], denial-of-service attacks [20], and false data injection attacks (FDIAs) [21], [22]. Hence, the associated control matters have drawn extensive attention owing to the FDIAs that destroy their integrity by injecting unexpected malicious information into the target systems. For example, [23] presented an event-driven adaptive nonfragile state estimation scheme for FO networked systems with cyberattacks. Afterward, an adaptive resilient control approach was provided related to FO nonlinear large-scale systems with actuator faults and deception attacks [7]. In addition, Zhao and Yang [24] explored the adaptive optimal control problem for large-scale nonlinear systems under unknown sensor FDIAs. However, noted that the reported control scheme in [24] cannot be directly applicable to FONSSs due to the complexity of fractional derivatives. In particular, since the original system states are destroyed by attack signals, only the compromised states can be applied to controller design. Therefore, studying the optimal secure control for FONSSs with FDIAs is an interesting and challenging task. Additionally, communication bandwidth limitation is a nonnegligible factor in control design, while most preceding control strategies are mainly implemented based on time-driven sampling, which leads to resource waste concerns.

Over the past few decades, event-triggered control (ETC) has attracted considerable interest for its ability to effectively reduce network congestion [25], [26], [27], [28], [29]. In [27], a ETC design with relative threshold was developed to reduce communication consumption for constrained parametric systems. Li et al. [28] exploited an event-triggered adaptive NN control for FONSSs. Indeed, during the execution of the ETC schemes, it is essential to constantly supervise the trigger threshold to ensure that the controller is updated properly. Whereas this inevitably entails some extra hardware support and increases resource occupation, which is not consistent with the original purpose. To hurdle this bottleneck, the adaptive self-triggered control (STC) scheme is proposed. In [30], the adaptive ST tracking control method was reported related

to nonlinear systems against input hysteresis. Nonetheless, it is regrettable that, to the authors' knowledge, the available results about STC for FONSSs is still very limited, which prompted our exploration in this paper. Furthermore, restricted by the physical environment and engineering requirements, the state constraint problems are worth taking into account in practice.

Hitherto, abundant meaningful findings have been reported for various state constraint issues, involving partial-state constraints [31], full-state constraints [32], bilateral constraints [33], and unilateral constraints [34]. More specifically, the barrier Lyapunov functions (BLFs) have been widely adopted to address state constraints. In [35] and [36], the BLF-aided adaptive control schemes were devised for FONSSs to ensure the constraint bounds were not transgressed. Nevertheless, the BLF-based constraint works inevitably require verifying a feasibility condition for the virtual controller. As a solution, [37] adopted a nonlinear mapping method to remove such restrictive conditions. Afterward, [38] provided the fuzzy optimal control scheme for state-constrained nonlinear systems where the feasibility condition was also canceled by employing a nonlinear function. However, since conducting fractional derivative entails a function to satisfy the convexity condition, the aforesaid methods are not suitable for FONSSs. Moreover, in some practical scenarios [39], [40], unilateral constraints tend to have stronger pertinence and higher control efficiency than bilateral constraints. To this end, the adaptive control algorithm based on a nonlinear function was proposed to achieve the unilateral full-state constraints (UFSCs) issue for FONSSs in [41]. Yet, how to develop an optimal control for constrained FONSSs to make a trade-off between system performance and control cost remains to be further studied.

Motivated by the foregoing discussions, an adaptive fuzzy optimized STC protocol is developed for constrained FONSSs with FDIAs in this study. The principal contributions are outlined as follows.

- 1) Distinguished from the existing literature [21], [22], [23], a coordinate change involving the unique nonlinear state-dependent function with convexity is formulated to alleviate the negative impacts of the FDIAs on system performance by integrating the compromised system and dynamic surface control technique. Particularly, the developed scheme successfully eliminates the feasibility conditions for the virtual controller existed in some BLFs-based constraint control methods [35], [36], [42]. Additionally, a nonlinear filter with the auxiliary term is presented to address the computational intricacy issue inherent in the conventional backstepping control approach [6], as opposed to the conventional first-order filter applied in [8].
- 2) Compared with the previous works [17], [24] and [41], the fractional HJB equation and the optimal solution can be derived by establishing an accessory system and equivalent transformation. Meantime, unlike integer-order RL using the gradient descent [14], [15], [16], the FO actor-critic weight learning laws are designed based on a Lyapunov function related to the optimal

NN weight error, which not only can ensure that the optimality of the NN weights can be achieved but also can simplify the form and design process while relaxing the continuous excitation conditions widely required in the traditional RL methods [12].

- 3) In contrast to the event-triggered mechanisms in [25], [27], [28], and [29], the utilized self-triggered mechanism can determine the next activation moment by calculating the current information, which make it possible that it is unessential to supervise continuously whether the trigger rule is satisfied. Hereby, the proposed optimal STC approach can reduce potential hardware requirements and save communication resources while minimizing the given performance index, which is substantial in practical applications.

The framework of this article is sketched as follows. Section II formulates model description and preliminaries. The detailed design procedure about adaptive optimized secure self-triggered controller and stability results are presented in Section III. Section IV provides a numerical example and an application example to demonstrate the validity and superiority of the developed control scheme. The whole work is concluded in Section V.

II. SYSTEM FORMULATION AND PRELIMINARIES

A. System Formulation

Consider the FONSSs with unknown FDIAs as follows:

$$\begin{cases} D^q x_i(t) = x_{i+1}(t) + f_i(\underline{x}_i) \\ D^q x_n(t) = u(t) + f_n(\underline{x}_n) \\ \bar{x}_i = (1 + \omega(t)) x_i \\ y(t) = x_1(t) \end{cases} \quad (1)$$

where $1 \leq i \leq n-1$, $q \in (0, 1)$ is the fractional order, $\underline{x}_i = [x_1, \dots, x_i]^T \in \mathbb{R}^i$ ($i = 1, \dots, n$), $u \in \mathbb{R}$ and $y \in \mathbb{R}$ represent the system state, input and output individually. $f_i(\underline{x}_i)$ denotes the unknown nonlinear term and $\bar{x}_i$ means the available compromised state vectors. Additionally, $\omega(t)$ stands for an unknown time-varying attack signal.

B. Fractional Calculus

Definition 1 [43]: The q th fractional integral of smooth function $g(t)$ is described as

$$I^q g(t) = \frac{1}{\Gamma(q)} \int_0^t \frac{g(\delta)}{(t-\delta)^{1-q}} d\delta$$

and the Caputo fractional derivative can be expressed as

$$D^q g(t) = \frac{1}{\Gamma(1-q)} \int_0^t \frac{g'(\delta)}{(t-\delta)^q} d\delta$$

in which $0 < q < 1 \in \mathbb{N}^+$, $\Gamma(q) = \int_0^\infty \delta^{q-1} e^{-\delta} d\delta$ denotes the Gamma function.

Definition 2 [44]: Define the Mittag-Leffler function as

$$E_{a,d}(\chi) = \sum_{k=0}^{\infty} \frac{\chi^k}{\Gamma(ka+d)}$$

where $a, d > 0$, $\chi \in \mathbb{C}$, and $E_{1,1}(\chi) = e^\chi$. Meanwhile, for $\alpha \in \mathbb{R}$, the Laplace transform can be expressed as

$$\mathcal{L}\{t^{d-1} E_{a,d}(-\alpha t^a)\} = \frac{s^{a-d}}{s^a + \alpha}.$$

Lemma 1 [43]: For $a \in (0, 2)$, $d \in \mathbb{R}$ and $c > 0$, there exists

$$|E_{a,d}(\chi)| \leq \frac{c}{1+|\chi|}, \quad |\chi| \geq 0$$

where $a\pi/2 < k < \min\{\pi, a\pi\}$ and $k \leq |\arg(\chi)| \leq \pi$.

Lemma 2 [18]: There exists a differential and convex function $g(v(t), t)$, then, for $\forall t > t_0$ and $0 < q \leq 1$, one obtains

$$D^q g(v(t), t) \leq \frac{\partial g(v(t), t)}{\partial v} D^q v(t).$$

Lemma 3 [44]: If there is a derivable function $g(t)$ and two constants $a, b > 0$, satisfying the following inequality:

$$D^q g(t) \leq -ag(t) + b, \quad t > t_0$$

then $g(t) \leq g(t_0)E_q(-at^q) + b/a$ for $q \in (0, 1]$.

Lemma 4 [45]: For $\forall \varpi > 0$ and $\Psi \in \mathbb{R}$, then

$$0 \leq |\Psi| - \Psi \tanh\left(\frac{\Psi}{\varpi}\right) \leq K\varpi, \quad K = 0.2785.$$

Assumption 1 [7]: The unknown time-varying function $\omega(t)$ meets $1 + \omega \neq 0$. Let $h = 1 + \omega$, and suppose there exist two unknown positive scales $\bar{h}$ and $\bar{h}_0$ satisfying $|h| \leq \bar{h}$, $|h^{-1} D^q h| \leq \bar{h}_0$.

Remark 1: For Assumption 1, $1 + \omega = 0$ reflects that the attack signals counteract the original state variables. In this case, the actual controller cannot be devised because the compromised state information is inaccessible. Besides, considering the physical constraints such as channel bandwidth and dynamic scope, it is realistic to assume that the attack signal and its FO derivative are bounded. Hence, such assumptions are basic and relatively moderate criteria for the controller design, and similar assumptions can be founded in the some existing results [21], [22], [24].

Control Objective: This work intends to present an optimized adaptive fuzzy secure STC algorithm for constrained FONSSs under FDIAs that ensures that all the signals in the closed-loop system (CLS) are bounded, and the stabilization errors tend toward a small region nearby the origin at the minimum costs.

C. Interval Type-2 Fuzzy Logic Systems

In this article, the IT2FLSs are employed to identify the unknown nonlinearities. Hypothesized that the IT2 fuzzy rule is established by the IF-THEN fuzzy rules as follows [46], [47].

$\mathbb{R}^m$: IF $\bar{s}_1$ is $\tilde{P}_1^m$ and $\dots$, and $\bar{s}_i$ is $\tilde{P}_i^m$ THEN y is $\tilde{Y}^m$, where $1 \leq m \leq K$, and K means the all rule number; $\bar{s}_1, \dots, \bar{s}_i$ and y denote the antecedent variables and consequent variable individually. $\tilde{P}_l^m = [y_l^m, y_r^m]$ represents the fuzzy sets with $1 \leq l \leq i$; y_l^m and y_r^m indicate crisp results.

$\tilde{P}_l^m$ with two membership functions $\underline{\mathfrak{S}}_{\tilde{P}_l^m}$ and $\bar{\mathfrak{S}}_{\tilde{P}_l^m}$ are expressed as

$$\underline{\mathfrak{S}}_{\tilde{P}_l^m} = \underline{c}_l^m \exp \left[-\frac{1}{2} \left(\frac{\bar{s}_l - a_l^m}{\underline{z}_l^m} \right)^2 \right]$$

$$\bar{\mathfrak{S}}_{\tilde{P}_l^m} = \bar{c}_l^m \exp \left[-\frac{1}{2} \left(\frac{\bar{s}_l - a_l^m}{\bar{z}_l^m} \right)^2 \right]$$

where $a_l^m, \underline{c}_l^m, \bar{c}_l^m, \underline{z}_l^m$ and $\bar{z}_l^m$ stand for the defined parameters satisfying $0 < \underline{c}_l^m \leq \bar{c}_l^m < 1$ and $\underline{z}_l^m < \bar{z}_l^m$.

The overall firing interval U^m of the m th rule $\mathbb{R}^m$ denotes $U^m(\bar{s}_i) = [\underline{Q}_m(\bar{s}_i), \bar{Q}_m(\bar{s}_i)]$, in which $\underline{Q}_m(\bar{s}_i) = \prod_{l=1}^i \underline{\mathfrak{S}}_{\tilde{P}_l^m}$ and $\bar{Q}_m(\bar{s}_i) = \prod_{l=1}^i \bar{\mathfrak{S}}_{\tilde{P}_l^m}$. Leveraging the mean values of $\underline{Q}_l$ and $\bar{Q}_l$, the total output of the IT2FLSs is

$$y = \frac{1}{2} \sum_{l=1}^K (\underline{Q}_l y_l^l + \bar{Q}_l y_l^l) = [W_l^T \ W_r^T] \begin{bmatrix} \frac{\varphi_l}{2} \\ \frac{\varphi_r}{2} \end{bmatrix} = W^T \varphi$$

where $W_l = [y_l^1, \dots, y_l^K]^T$, $W_r = [y_r^1, \dots, y_r^K]^T$, $\varphi_l = [\underline{Q}_l, \dots, \bar{Q}_l]^T$, and $\varphi_r = [\bar{Q}_l, \dots, \underline{Q}_l]^T$.

Lemma 5 ([46], [47]): The unknown smooth function $F(\bar{s}_i)$ over a bounded compact set Ω_0 can be identified through the following IT2FLS:

$$\sup_{\bar{s}_i \in \Omega_0} |F(\bar{s}_i) - W^T \varphi(\bar{s}_i)| \leq \varepsilon(\bar{s}_i)$$

where $\varepsilon(\bar{s}_i)$ stands for the approximation error with the supreme $\bar{\varepsilon} > 0$.

III. MAIN RESULTS

This segment will fulfill the design of an adaptive fuzzy optimal secure control scheme with self-triggered mechanism and stability analysis for constrained FONSS (1) via RL and recursive control technique.

A. Controller Design

Considering the FONSS (1) being subjected to the FDIAs, the available compromised system states are used in the subsequent controller design. Taking the q th derivative of $\bar{x}_i$, one has

$$D^q \bar{x}_i \leq x_i D^q h + h D^q x_i. \quad (2)$$

From (2), there must exist a nonnegative time-varying function $p_i(t)$ satisfying

$$D^q \bar{x}_i = x_i D^q h + h D^q x_i - p_i(t). \quad (3)$$

Furthermore, the compromised system is rephrased as

$$\begin{cases} D^q \bar{x}_i = \bar{x}_{i+1} + \bar{x}_i h^{-1} D^q h + h f_i - p_i \\ D^q \bar{x}_n = hu + \bar{x}_n h^{-1} D^q h + h f_n - p_n \\ \bar{y} = \bar{x}_1. \end{cases} \quad (4)$$

Meanwhile, to fulfill the UFSC, a nonlinear state-dependent function with $1 \leq i \leq n$ is constructed as

$$\lambda_i(t) = \frac{\bar{x}_i}{c_i - \bar{x}_i} \quad (5)$$

where c_i is the design scale.

Remark 2: From (5), it can be observed that $\lambda_i(t)$ entirely depends on $\bar{x}_i(t)$, and for any initial condition $\bar{x}_i(0) < c_i$ or $\bar{x}_i(0) > c_i$, $\lambda_i(t) \rightarrow \infty$ if and only if $\bar{x}_i \rightarrow c_i$. This reveals that the constraint exerted on $\bar{x}_i(t)$ can be satisfied as long as $\lambda_i(t)$ remains bounded. Consequently, the issue of imposing the UFSC on $\bar{x}_i(t)$ is converted into assuring the boundedness of λ_i . Besides, by setting c_i to be either positive or negative, one can obtain a positive upper-bound or negative lower-bound constraint, respectively. Additionally, it should be noted that the similar UFSC effects can be achieved by altering the form to $\bar{x}_i/(\bar{x}_i - c_i)$. Given the similarity in treatment, the functional form $\bar{x}_i/(c_i - \bar{x}_i)$ is adopted in this work for subsequent discussion.

Going ahead, one can deduce that

$$D^q \lambda_i \leq \zeta_{\lambda_i} D^q \bar{x}_i, \quad \bar{x}_i = \zeta_{\bar{x}_i} \lambda_i \quad (6)$$

where $\zeta_{\lambda_i} = c_i/(c_i - \bar{x}_i)^2$ and $\zeta_{\bar{x}_i} = c_i - \bar{x}_i$.

Then, similar to (2) and there exists $q_i(t) \geq 0$, one can get

$$\begin{cases} D^q \lambda_i = \zeta_{\lambda_i} (\bar{x}_{i+1} + \mu_i^* \bar{x}_i + h f_i - p_i) - q_i \\ D^q \lambda_n = \zeta_{\lambda_n} (hu + \mu_n^* \bar{x}_n + h f_n - p_n) - q_n \end{cases} \quad (7)$$

where $h^{-1} D^q h = \mu^*$.

The coordinate transformation is defined as

$$\begin{cases} z_1 = \lambda_1 \\ z_i = \lambda_i - \eta_i \\ \xi_i = \eta_i - \alpha_{i-1}, \quad i = 2, \dots, n \end{cases} \quad (8)$$

in which z_1, z_i , and ξ_i represent the surface errors and the filtered error, respectively. η_i denotes the filtered signal generated by passing the virtual control function α_{i-1} through the following FO filter with $\eta_i(0) = \alpha_{i-1}(0)$:

$$D^q \eta_i = -\hat{\Lambda}_i \tanh \left(\frac{\hat{\Lambda}_i \xi_i}{\varpi_i} \right) - r_i \xi_i \quad (9)$$

where $\varpi_i, r_i > 0$ are design parameters and $\hat{\Lambda}_i$ is the estimated scale of Λ_i and its definition will be provided later.

Remark 3: Affected by malicious attacks, the original state information are no longer suitable for feedback control design. In this situation, only the state variables confused by attack signal can be exploited to design controller. Thereby, a coordinate transformation based on the damaged system is proposed to address the unknown FDIAs. Meantime, to further relieve the influences of the attacks, the nonlinear state-dependent function is formulated to realize the UFSCs. Notably, the original state variables are not involved in the virtual controller and actual controller during the recursive design procedure, which are merely used for theoretical analysis.

The optimal performance index is selected as

$$J^*(z(t)) = \min_{\alpha \in \Omega_s} \int_t^\infty (z^2(\tau) + \alpha^2) d\tau \quad (10)$$

where Ω_s is the set of all admissible control inputs. $z^2(\tau) + \alpha^2$ denotes the cost function to be minimized and α is the virtual controller. According to the Bellman's optimality theory, yields the HJB equation as

$$H \left(z, \alpha^*, \frac{\partial J^*}{\partial z} \right) = z^2 + \alpha^{*2} + \frac{\partial J^*}{\partial z} \dot{z} = 0. \quad (11)$$

Define the auxiliary variable $e = D^{q-1}z \in \mathbb{R}^n$. Based on Property 2 in [18], it can be derived as

$$\dot{e} = D^1 D^{q-1}z = D^q z. \quad (12)$$

and $z = D^{1-q} D^{q-1}z = D^{1-q}e$.

Then, (10) and (11) can be rewritten as

$$J^*(D^{1-q}e) = \int_t^\infty \left((D^{1-q}e)^2 + \alpha^{*2} \right) d\tau \quad (13)$$

$$H(z, \alpha^*, \nabla J_e^*) = (D^{1-q}e)^2 + \alpha^{*2} + \nabla J_e^{*T} \dot{e} = 0 \quad (14)$$

where $\nabla J_e^* = \partial J^*/\partial e$. Let $\partial H(z, \alpha^*, \nabla J_e^*)/\partial \alpha^* = 0$, the optimal virtual controller can be obtained as

$$\alpha^* = -\frac{\zeta_\lambda}{2} \nabla J_e^*. \quad (15)$$

Moreover, since $z = D^{1-q}e$, $\dot{e} = D^q z$ and $J^*(D^{1-q}e) = J^*(z)$, one can get the fractional HJB as

$$H(z, \alpha^*, \nabla J_z^*) = z^2 + \alpha^{*2} + \nabla J_z^{*T} D^q z = 0 \quad (16)$$

where $\nabla J_z^* = \partial J^*/\partial (D^{q-1}z)$. Then, the optimal virtual controller is

$$\alpha^* = -\frac{\zeta_\lambda}{2} \nabla J_z^*. \quad (17)$$

Evidently, the HJB (14) and (16) are equivalent and the optimal controller (15) is also equivalent to (17).

Remark 4: From Lemma 2, it is obtained that the FO chain rule is unapplicable and the fractional HJB equation cannot be derived directly. To this end, the fractional dynamics are introduced to the HJB equation by drawing support from the auxiliary system (12), and the fractional HJB (16) together with the optimal controller (17) can be deduced. Then, the above equivalent relations indirectly show that they are plausible. Besides, if the RL strategy is straightforwardly devised for the accessory system (12), it will make it more difficult to conducting the stability analysis. Therefore, deriving the fractional HJB equation is imperative.

Step 1: Computing the FO derivative of z_1 , it produces

$$D^q z_1 = \zeta_{\lambda_1} (\bar{x}_2 + \mu_1^* \bar{x}_1 + hf_1 - p_1) - q_1. \quad (18)$$

Regard $\bar{x}_2$ as the optimal virtual control α_1^* and α_1 is the virtual control input. Then, the optimal performance index is specified as

$$J_1^*(z_1(t)) = \min_{\alpha_1 \in \Omega_1} \int_t^\infty \left(z_1^2(\tau) + \alpha_1^2(z_1) \right) d\tau.$$

From the above, the fractional HJB equation is described as

$$H_1 = z_1^2 + \alpha_1^{*2} + \nabla J_{z_1}^{*T} D^q z_1 = 0$$

where $\nabla J_{z_1}^* = \partial J^*/\partial (D^{q-1}z_1)$. By solving $\partial H_1/\partial \alpha_1^* = 0$, the optimal virtual controller is

$$\alpha_1^* = -\frac{\zeta_{\lambda_1}}{2} \nabla J_{z_1}^*.$$

To reach the optimal control target, constructing the auxiliary cost function $J_1^0 = \zeta_{\lambda_1} \nabla J_{z_1}^* - 2k_1 z_1/(\zeta_{\lambda_1} \zeta_{\bar{x}_2}) - 2(\mu_1^* \bar{x}_1 + \theta_1^{*T} \phi_1 + \varepsilon_1 + \zeta_{\lambda_1} z_1/2)/\zeta_{\bar{x}_2} - 5\zeta_{\lambda_1} \zeta_{\bar{x}_2} z_1/2$, where $k_1 > 0$ is the design parameter. Meanwhile, the IT2FLS is used to

model the unknown item $F_1(\check{x}_1) = hf_1 - p_1 - q_1/\zeta_{\lambda_1} = \theta_1^{*T} \phi_1(\check{x}_1) + \varepsilon_1(\check{x}_1)$, where $\varepsilon_1(\check{x}_1)$ is the approximation error satisfying $|\varepsilon_1(\check{x}_1)| \leq \bar{\varepsilon}_1$ with $\bar{\varepsilon}_1 > 0$ and the input $\check{x}_1 = \bar{x}_1$.

Then, we can get

$$\alpha_1^* = -\frac{1}{\zeta_{\lambda_1} \zeta_{\bar{x}_2}} k_1 z_1 - \frac{1}{\zeta_{\bar{x}_2}} \left(\mu_1^* \bar{x}_1 + \theta_1^{*T} \phi_1 + \varepsilon_1 + \frac{1}{2} \zeta_{\lambda_1} z_1 \right) - \frac{5}{4} \zeta_{\lambda_1} \zeta_{\bar{x}_2} z_1 - \frac{1}{2} J_1^0 \quad (19)$$

$$\nabla J_{z_1}^* = \frac{2}{\zeta_{\lambda_1} \zeta_{\bar{x}_2}} k_1 z_1 + \frac{2}{\zeta_{\lambda_1} \zeta_{\bar{x}_2}} \left(\mu_1^* \bar{x}_1 + \theta_1^{*T} \phi_1 + \varepsilon_1 + \frac{1}{2} \zeta_{\lambda_1} z_1 \right) + \frac{5}{2} \zeta_{\bar{x}_2} z_1 + \frac{1}{\zeta_{\lambda_1}} J_1^0. \quad (20)$$

Using an NN to approximate the unknown term J_1^0 yields

$$J_1^0 = W_1^{*T} \varphi_1(\check{z}_1) + \varepsilon_1(\check{z}_1) \quad (21)$$

where W_1^{*T} and ε_1 represent the ideal weight vector and approximation error, correspondingly. φ_1 denotes the basis function with the input $\check{z}_1 = [\bar{x}_1, \bar{x}_2, z_1]^T$. Since the ideal parameters are unobtainable, an actor-critic NN is conducted to train the cost function J_1^0 online. One gets

$$\hat{\alpha}_1 = -\frac{1}{\zeta_{\lambda_1} \zeta_{\bar{x}_2}} k_1 z_1 - \frac{1}{\zeta_{\bar{x}_2}} \left(\hat{\mu}_1 \bar{x}_1 + \hat{\theta}_1^T \phi_1 + \frac{\zeta_{\lambda_1} z_1}{2} \right) - \frac{5}{4} \zeta_{\lambda_1} \zeta_{\bar{x}_2} z_1 - \frac{1}{2} \hat{W}_{a1}^T \varphi_1 \quad (22)$$

$$\nabla \hat{J}_{z_1} = \frac{2}{\zeta_{\lambda_1} \zeta_{\bar{x}_2}} k_1 z_1 + \frac{2}{\zeta_{\lambda_1} \zeta_{\bar{x}_2}} \left(\hat{\mu}_1 \bar{x}_1 + \hat{\theta}_1^T \phi_1 + \frac{\zeta_{\lambda_1} z_1}{2} \right) + \frac{5}{2} \zeta_{\bar{x}_2} z_1 + \frac{1}{\zeta_{\lambda_1}} \hat{W}_{c1}^T \varphi_1 \quad (23)$$

where $\hat{\mu}_1, \hat{\theta}_1, \hat{W}_{a1}$ and $\hat{W}_{c1}$ indicate the estimated value of $\mu_1^*, \theta_1^*, W_{a1}^*$ and W_{c1}^* , correspondingly.

Thereafter, substituting (22) and (23) into the fractional HJB equation produces an approximation error as

$$\begin{aligned} E_1 &= \hat{H}_1(z_1, \hat{\alpha}_1, \nabla \hat{J}_{z_1}) - H_1(z_1, \alpha_1^*, \nabla J_{z_1}^*) \\ &= z_1^2 + \left[\frac{1}{\zeta_{\lambda_1} \zeta_{\bar{x}_2}} k_1 z_1 + \frac{1}{\zeta_{\bar{x}_2}} \left(\hat{\theta}_1^T \phi_1 + \frac{\zeta_{\lambda_1} z_1}{2} \right) + \frac{1}{2} \hat{W}_{a1}^T \varphi_1 + \frac{\hat{\mu}_1 \bar{x}_1}{\zeta_{\bar{x}_2}} + \frac{5}{4} \zeta_{\lambda_1} \zeta_{\bar{x}_2} z_1 \right]^2 \\ &\quad + \left[\frac{2k_1 z_1}{\zeta_{\lambda_1}^2 \zeta_{\bar{x}_2}} + \frac{\hat{W}_{c1}^T \varphi_1}{\zeta_{\lambda_1}} + \frac{5\zeta_{\bar{x}_2} z_1}{2} + \frac{2}{\zeta_{\lambda_1} \zeta_{\bar{x}_2}} \left(\hat{\mu}_1 \bar{x}_1 + \hat{\theta}_1^T \phi_1 + \frac{\zeta_{\lambda_1} z_1}{2} \right) \right]^T \left[-\frac{1}{\zeta_{\bar{x}_2}} k_1 z_1 - \frac{\zeta_{\lambda_1}}{\zeta_{\bar{x}_2}} \left(\hat{\mu}_1 \bar{x}_1 + \hat{\theta}_1^T \phi_1 + \frac{\zeta_{\lambda_1} z_1}{2} \right) - \frac{5}{4} \zeta_{\lambda_1}^2 \zeta_{\bar{x}_2} z_1 + \zeta_{\lambda_1} \left(-\frac{1}{2} \hat{W}_{a1}^T \varphi_1 + \mu_1^* \bar{x}_1 + \theta_1^{*T} \phi_1 + \varepsilon_1 \right) \right]. \end{aligned}$$

Obviously, $\hat{\alpha}_1$ is expected to guarantee $E_1 \rightarrow 0$. Since the optimal solution is unique, $\hat{H}_1(z_1, \hat{\alpha}_1, \nabla \hat{J}_{z_1}) = 0$ holds and the following equation also holds:

$$\frac{\partial E_1}{\partial \hat{W}_{a1}} = \frac{1}{2} \varphi_1 \varphi_1^T (\hat{W}_{a1} - \hat{W}_{c1}) = 0. \quad (24)$$

When $\hat{W}_{a1} - \hat{W}_{c1} = 0$, then (24) is satisfied. Thus, the following Lyapunov function is constructed:

$$S_1 = (\hat{W}_{a1} - \hat{W}_{c1})^T (\hat{W}_{a1} - \hat{W}_{c1}).$$

The q th time derivative of S_1 is

$$D^q S_1 \leq \frac{\partial S_1}{\partial \hat{W}_{a1}} D^q \hat{W}_{a1} + \frac{\partial S_1}{\partial \hat{W}_{c1}} D^q \hat{W}_{c1} \quad (25)$$

where $\partial S_1 / \partial \hat{W}_{a1} = -\partial S_1 / \partial \hat{W}_{c1} = 2(\hat{W}_{a1} - \hat{W}_{c1})$. The actor-critic-identifier weight learning laws are designed as

$$D^q \hat{W}_{a1} = -\varphi_1 \varphi_1^T [\gamma_{a1} (\hat{W}_{a1} - \hat{W}_{c1}) + \gamma_{c1} \hat{W}_{c1}] \quad (26)$$

$$D^q \hat{W}_{c1} = -\gamma_{c1} \varphi_1 \varphi_1^T \hat{W}_{c1} \quad (27)$$

$$D^q \hat{\theta}_1 = z_1 \zeta_{\lambda_1} \phi_1 - \rho_1 \hat{\theta}_1 \quad (28)$$

where γ_{a1} , γ_{c1} , and ρ_1 are positive constants. Introducing (26) and (27) into (25), one can get

$$D^q S_1 \leq -\frac{1}{2} \frac{\partial S_1^T}{\partial \hat{W}_{a1}} \varphi_1 \varphi_1^T \frac{\partial S_1}{\partial \hat{W}_{a1}} \leq 0.$$

Therefore, it can be obtained that $\hat{W}_{a1} - \hat{W}_{c1}$ is asymptotically stable. Additionally, the adaptive law is designed as

$$D^q \hat{\mu}_1 = z_1 \zeta_{\lambda_1} \bar{x}_1 - \sigma_1 \hat{\mu}_1 \quad (29)$$

where $\sigma_1 > 0$ is a design parameter.

Consider the Lyapunov function: $V_1 = (1/2)z_1^2 + (1/2)\tilde{\mu}_1^2 + (1/2)\tilde{\theta}_1^T \tilde{\theta}_1 + (1/2)\tilde{W}_{a1}^T \tilde{W}_{a1} + (1/2)\tilde{W}_{c1}^T \tilde{W}_{c1}$. Let $L_1 = (1/2)z_1^2 + (1/2)\tilde{\mu}_1^2 + (1/2)\tilde{\theta}_1^T \tilde{\theta}_1$ and $\bar{L}_1 = (1/2)\tilde{W}_{a1}^T \tilde{W}_{a1} + (1/2)\tilde{W}_{c1}^T \tilde{W}_{c1}$. According to (26)–(29), taking the fractional derivative of L_1 produces

$$\begin{aligned} D^q L_1 \leq & -k_1 z_1^2 + z_1 \zeta_{\lambda_1} \left[-\tilde{\mu}_1 \bar{x}_1 - \tilde{\theta}_1^T \phi_1 - \frac{\zeta_{\lambda_1} z_1}{2} + \varepsilon_1 \right. \\ & + \zeta_{x_2} (z_2 + \xi_2) - \frac{1}{2} \zeta_{x_2} \hat{W}_{a1}^T \varphi_1 \left. \right] - \frac{5}{4} \zeta_{\lambda_1}^2 \zeta_{x_2}^2 z_1^2 \\ & + z_1 \zeta_{\lambda_1} (\tilde{\mu}_1 \bar{x}_1 + \tilde{\theta}_1^T \phi_1) - \sigma_1 \tilde{\mu}_1 \hat{\mu}_1 - \rho_1 \tilde{\theta}_1^T \hat{\theta}_1 \end{aligned} \quad (30)$$

where $\tilde{\mu}_1 = \hat{\mu}_1 - \mu_1^*$ and $\tilde{\theta}_1 = \hat{\theta}_1 - \theta_1^*$.

Then, utilizing Young's inequality yields

$$\begin{cases} \zeta_{\lambda_1} \zeta_{x_2} z_1 (z_2 + \xi_2) \leq \zeta_{\lambda_1}^2 \zeta_{x_2}^2 z_1^2 + \frac{1}{2} z_2^2 + \frac{\xi_2^2}{2} \\ -\zeta_{\lambda_1} \zeta_{x_2} z_1 \hat{W}_{a1}^T \varphi_1 \leq \frac{\zeta_{\lambda_1}^2 \zeta_{x_2}^2 z_1^2}{2} + \frac{1}{2} (\hat{W}_{a1}^T \varphi_1)^2 \\ z_1 \zeta_{\lambda_1} \varepsilon_1 \leq \frac{z_1^2 \zeta_{\lambda_1}^2}{2} + \frac{\varepsilon_1^2}{2}. \end{cases} \quad (31)$$

Substituting (31) into (30), one can get

$$\begin{aligned} D^q L_1 \leq & -k_1 z_1^2 + \frac{1}{4} (\hat{W}_{a1}^T \varphi_1)^2 + \frac{\xi_2^2}{2} + \frac{1}{2} z_2^2 \\ & + \frac{\varepsilon_1^2}{2} - \sigma_1 \tilde{\mu}_1 \hat{\mu}_1 - \rho_1 \tilde{\theta}_1^T \hat{\theta}_1. \end{aligned} \quad (32)$$

Remark 5: Based on the earlier discussion, the analytical solution of the fractional HJB equation H_1 denotes the optimal control. Yet, it is a fact that $\nabla J_{z_1}^* = \partial J^* / \partial (D^{q-1} z_1)$ and α_1^* cannot be directly obtained. Thereby, the actor-critic

NNs are introduced to form $\nabla \hat{J}_{z_1}$ and $\hat{\alpha}_1$, and the identifier IT2FLS is employed to model the unknown nonlinear item $F_1(\bar{x}_1)$, respectively. Noted that these approximation terms will inevitably yield an approximation error E_1 , which impels the NN weights $\hat{W}_{a1}$ and $\hat{W}_{c1}$ to be trained online until E_1 is minimized complying with the basic principle of the RL and actor-critic architecture. Additionally, the continuous excitation condition is not needed as discussed in [12] since the weights are trained to the optimum via the Lyapunov function scheme formulated in (24)–(27).

Step i ($i = 2, \dots, n-1$): The differentiation of z_i can be expressed as

$$D^q z_i = \zeta_{\lambda_i} (\bar{x}_{i+1} + \mu_i^* \bar{x}_i + h f_i - p_i) - q_i - D^q \eta_i. \quad (33)$$

Similarly, define the optimal performance index as

$$J_i^*(z_i(t)) = \min_{\alpha_i \in \Omega_{\alpha_i}} \int_t^\infty (z_i^2(\tau) + \alpha_i^2(z_i)) d\tau.$$

The fractional HJB equation is

$$H_i = z_i^2 + \alpha_i^{*2} + \nabla J_{z_i}^{*T} D^q z_i$$

where $\nabla J_{z_i}^* = \partial J^* / \partial (D^{q-1} z_i)$. Let $\partial H_i / \partial \alpha_i^* = 0$, the optimal virtual controller is obtained as

$$\alpha_i^* = -\frac{1}{2} \zeta_{\lambda_i} \nabla J_{z_i}^*.$$

Likewise, $J_i^0 = -2(k_i z_i - D^q \eta_i) / (\zeta_{\lambda_i} \zeta_{\bar{x}_{i+1}}) - 2(\mu_i^* \bar{x}_i + \theta_i^* \phi_i + \zeta_{\lambda_i} z_i / 2 + \varepsilon_i) / \zeta_{\bar{x}_{i+1}} - 5\zeta_{\lambda_i} \zeta_{\bar{x}_{i+1}} z_i / 2 + \zeta_{\lambda_i} \nabla J_{z_i}^*$, where $k_i > 0$ is the defined parameter. Simultaneously, the IT2FLS is adopted to approximate the unknown term $F_i(\bar{x}_i) = h f_i - p_i - q_i / \zeta_{\lambda_i} = \theta_i^{*T} \phi_i(\bar{x}_i) + \varepsilon_i(\bar{x}_i)$, where $\varepsilon_i(\bar{x}_i)$ denotes the approximation error with $|\varepsilon_i(\bar{x}_i)| \leq \bar{\varepsilon}_i$ and the input $\bar{x}_i = \bar{x}_i$.

By utilizing an NN to approximate J_i^0 leads to

$$J_i^0 = W_i^{*T} \varphi_i(\bar{z}_i) + \varepsilon_i(\bar{z}_i) \quad (34)$$

where W_i^{*T} and ε_i denote the ideal weight vector and approximation error, respectively, and φ_i indicates the basis function with the input vector $\bar{z}_i = [\bar{x}_i, \bar{x}_{i+1}, z_i, D^q \eta_i]^T$. The actor-critic NN is applied to train J_i^0 , one yields

$$\begin{aligned} \hat{\alpha}_i = & -\frac{1}{\zeta_{\lambda_i} \zeta_{\bar{x}_{i+1}}} (k_i z_i - D^q \eta_i) \\ & - \frac{1}{\zeta_{\bar{x}_{i+1}}} \left(\hat{\mu}_i \bar{x}_i + \hat{\theta}_i^T \phi_i + \frac{\zeta_{\lambda_i}}{2} z_i \right) - \frac{5}{4} \zeta_{\lambda_i} \zeta_{\bar{x}_{i+1}} z_i \\ & - \frac{1}{2} \hat{W}_{ai}^T \varphi_i \end{aligned} \quad (35)$$

$$\begin{aligned} \nabla \hat{J}_{z_i} = & \frac{2}{\zeta_{\lambda_i}^2 \zeta_{\bar{x}_{i+1}}} (k_i z_i - D^q \eta_i) \\ & + \frac{2}{\zeta_{\lambda_i} \zeta_{\bar{x}_{i+1}}} \left(\hat{\mu}_i \bar{x}_i + \hat{\theta}_i^T \phi_i + \frac{\zeta_{\lambda_i}}{2} z_i \right) + \frac{5}{2} \zeta_{\bar{x}_{i+1}} z_i \\ & + \frac{1}{\zeta_{\lambda_i}} \hat{W}_{ci}^T \varphi_i \end{aligned} \quad (36)$$

where $\hat{\mu}_i$, $\hat{\theta}_i$, $\hat{W}_{ai}$ and $\hat{W}_{ci}$ stand for the estimated value of μ_i^* , θ_i^* , W_{ai}^* and W_{ci}^* , respectively. Similar to the analysis in step 1, the weight learning laws are proposed as

$$D^q \hat{W}_{ai} = -\varphi_i \varphi_i^T [\gamma_{ai} (\hat{W}_{ai} - \hat{W}_{ci}) + \gamma_{ci} \hat{W}_{ci}] \quad (37)$$

$$D^q \hat{W}_{ci} = -\gamma_{ci} \varphi_i \varphi_i^T \hat{W}_{ci} \quad (38)$$

where $\gamma_{ai}, \gamma_{ci} > 0$. In addition, design the identifier weight learning law and parameter updating law as

$$D^q \hat{\theta}_i = z_i \zeta_{\lambda_i} \phi_i - \rho_i \hat{\theta}_i \quad (39)$$

$$D^q \hat{\mu}_i = z_i \zeta_{\lambda_i} \bar{x}_i - \sigma_i \hat{\mu}_i \quad (40)$$

$$D^q \hat{\Lambda}_i = |\xi_i| - \ell_i \hat{\Lambda}_i \quad (41)$$

where ρ_i, σ_i and ℓ_i are the positive defined constants.

Subsequently, the Lyapunov function is defined as $V_i = \frac{1}{2} z_i^2 + (1/2) \tilde{\mu}_i^2 + (1/2) \xi_i^2 + (1/2) \tilde{\Lambda}_i^2 + (1/2) \tilde{\theta}_i^T \tilde{\theta}_i + (1/2) \tilde{W}_{ai}^T \tilde{W}_{ai} + (1/2) \tilde{W}_{ci}^T \tilde{W}_{ci}$.

Based on (8) and (9), one gets

$$D^q \xi_i = -\hat{\Lambda}_i \tanh\left(\frac{\hat{\Lambda}_i \xi_i}{\varpi_i}\right) - r_i \xi_i + \tilde{\Lambda}_i \quad (42)$$

where $|\tilde{\Lambda}_i| = | -D^q \alpha_{i-1}| \leq \Lambda_i$ with $\Lambda_i > 0$ and $\tilde{\Lambda}_i = \hat{\Lambda}_i - \Lambda_i$.

Let $L_i = (1/2) z_i^2 + (1/2) \tilde{\mu}_i^2 + (1/2) \xi_i^2 + (1/2) \tilde{\Lambda}_i^2 + (1/2) \tilde{\theta}_i^T \tilde{\theta}_i$ and $\tilde{L}_i = (1/2) \tilde{W}_{ai}^T \tilde{W}_{ai} + 1/2 \tilde{W}_{ci}^T \tilde{W}_{ci}$, one can obtain

$$\begin{aligned} D^q L_i &\leq -k_i z_i^2 + z_i \zeta_{\lambda_i} \left[-\tilde{\mu}_i \bar{x}_i - \tilde{\theta}_i^T \phi + \varepsilon_i - \frac{\zeta_{\lambda_{i+1}}}{2} \hat{W}_{ai}^T \varphi_i \right. \\ &\quad \left. - \frac{\zeta_{\lambda_i} z_i}{2} + \zeta_{\lambda_{i+1}} (z_{i+1} + \xi_{i+1}) \right] - \frac{5}{4} \zeta_{\lambda_i}^2 \zeta_{\lambda_{i+1}}^2 z_i^2 \\ &\quad + z_i \zeta_{\lambda_i} (\tilde{\mu}_i \bar{x}_i + \tilde{\theta}_i^T \phi_i) - \hat{\Lambda}_i \xi_i \tanh\left(\frac{\hat{\Lambda}_i \xi_i}{\varpi_i}\right) - r_i \xi_i^2 \\ &\quad + |\xi_i| (\Lambda_i + \tilde{\Lambda}_i) - \sigma_i \tilde{\mu}_i \hat{\mu}_i - \ell_i \tilde{\Lambda}_i \hat{\Lambda}_i - \rho_i \tilde{\theta}_i^T \hat{\theta}_i \end{aligned} \quad (43)$$

where $\tilde{\theta}_i = \hat{\theta}_i - \theta_i^*$ and $\tilde{\mu}_i = \hat{\mu}_i - \mu_i^*$.

Following Lemma 4 and Young's inequality, it achieves

$$\begin{aligned} D^q L_i &\leq -k_i z_i^2 + \frac{1}{4} (\hat{W}_{ai}^T \varphi_i)^2 + \frac{\xi_{i+1}^2}{2} + \frac{z_{i+1}^2}{2} + \frac{\tilde{\varepsilon}_i^2}{2} \\ &\quad - r_i \xi_i^2 - \sigma_i \tilde{\mu}_i \hat{\mu}_i - \ell_i \tilde{\Lambda}_i \hat{\Lambda}_i - \rho_i \tilde{\theta}_i^T \hat{\theta}_i + K \varpi_i. \end{aligned} \quad (44)$$

Step n: Recalling the definition of z_n in (8), we have

$$D^q z_n = \zeta_{\lambda_n} (hu + \mu_n^* \bar{x}_n + hf_n - p_n) - q_n - D^q \eta_n. \quad (45)$$

The optimal performance function in this step is defined as

$$J_n^*(z_n(t)) = \min_{u \in \Omega_s} \int_t^\infty (z_n^2(\tau) + u^2(z_n)) d\tau.$$

The fractional HJB equation with respect to u^* is

$$H_n = z_n^2 + u^{*2} + \nabla J_{z_n}^{*T} D^q z_n$$

where $\nabla J_{z_n}^* = \partial J^* / \partial (D^{q-1} z_n)$. By calculating $\partial H_n / \partial u^* = 0$, the optimal controller is attained as

$$u^* = -\frac{1}{2} h \zeta_{\lambda_n} \nabla J_{z_n}^*.$$

Let $J_n^0 = h \zeta_{\lambda_n} \nabla J_{z_n}^* - 2(k_n z_n - D^q \eta_n) / (h \zeta_{\lambda_n}) - 2(\mu_n^* \bar{x}_n + \theta_n^{*T} \phi_n + \varepsilon_n + \zeta_{\lambda_n} z_n / 2) / h - h \zeta_{\lambda_n} z_n / 2$, where $k_n > 0$ is the defined constant. Herein, the IT2FLS is employed to model the inaccessible function $F_i(\check{x}_n) = hf_n - p_n - q_n / \zeta_{\lambda_n} = \theta_n^{*T} \phi_n(\check{x}_n) + \varepsilon_n(\check{x}_n)$, where $\varepsilon_n(\check{x}_n)$ denotes the approximation error with $|\varepsilon_n(\check{x}_n)| \leq \bar{\varepsilon}_n$ and the input $\check{x}_n = \bar{x}_n$.

Applying an NN to identify J_n^0 , it produces

$$J_n^0 = W_n^{*T} \varphi_n(\check{z}_n) + \varepsilon_n(\check{z}_n) \quad (46)$$

in which W_n^{*T} expresses the ideal weight vector, ε_n is the approximation error meeting $|\varepsilon_n| \leq \bar{\varepsilon}_n$, φ_n means the basis function with $\check{z}_n = [\bar{x}_n, z_n, D^q \eta_n]^T$. Similarly, using an actor-critic NN to train J_n^0 , yielding the approximate optimal controller $\hat{u}$ and $\nabla \hat{J}_{z_n}$

$$\begin{aligned} \hat{u} &= -\frac{1}{h \zeta_{\lambda_n}} (k_n z_n - D^q \eta_n) - \frac{1}{h} \left(\hat{\mu}_n \bar{x}_n + \hat{\theta}_n^T \phi_n + \frac{\zeta_{\lambda_n}}{2} z_n \right) \\ &\quad - \frac{h \zeta_{\lambda_n}}{4} z_n - \frac{1}{2} \hat{W}_{an}^T \varphi_n \end{aligned} \quad (47)$$

$$\begin{aligned} \nabla \hat{J}_{z_n} &= \frac{2}{h^2 \zeta_{\lambda_n}^2} (k_n z_n - D^q \eta_n) + \frac{2}{h^2 \zeta_{\lambda_n}} \left(\hat{\mu}_n \bar{x}_n + \hat{\theta}_n^T \phi_n + \frac{\zeta_{\lambda_n}}{2} z_n \right) \\ &\quad + \frac{1}{2} z_n + \frac{1}{h \zeta_{\lambda_n}} \hat{W}_{cn}^T \varphi_n \end{aligned} \quad (48)$$

where $\hat{\mu}_n, \hat{\theta}_n, \hat{W}_{an}$ and $\hat{W}_{cn}$ represent the estimated value of $\mu_n^*, \theta_n^*, W_{an}^*$ and W_{cn}^* , respectively. Referring to the previous steps, the actor-critic-identifier weight learning laws and adaptive laws are specified as

$$D^q \hat{W}_{an} = -\varphi_n \varphi_n^T \left[\gamma_{an} (\hat{W}_{an} - \hat{W}_{cn}) + \gamma_{cn} \hat{W}_{cn} \right] \quad (49)$$

$$D^q \hat{W}_{cn} = -\gamma_{cn} \varphi_n \varphi_n^T \hat{W}_{cn} \quad (50)$$

$$D^q \hat{\theta}_n = z_n \zeta_{\lambda_n} \phi_n - \rho_n \hat{\theta}_n \quad (51)$$

$$D^q \hat{\mu}_n = z_n \zeta_{\lambda_n} \bar{x}_n - \sigma_n \hat{\mu}_n \quad (52)$$

$$D^q \hat{\Lambda}_n = |\xi_n| - \ell_n \hat{\Lambda}_n \quad (53)$$

where $\gamma_{an}, \gamma_{cn}, \rho_n, \sigma_n$ and ℓ_n are positive design scales.

Subsequently, the self-triggered mechanism is utilized to reduce the communication burden, which is formulated as

$$u(t) = \vartheta(t_k), \quad \forall t \in [t_k, t_{k+1}) \quad (54)$$

$$t_{k+1} = t_k + \frac{\beta |u(t)| + M}{\max\{l, |\dot{u}(t)|\}} \quad (55)$$

where $0 < \beta < 1, M, l > 0$. t_k and t_{k+1} denote the current and next update time with $t_k < t_{k+1}, k \in \mathbb{N}^+$. $\beta |u(t)| + M$ means the control signal gap between two consecutive trigger times. l and $|\dot{u}(t)|$ are the adjustment rates of the gap

$$\vartheta = -(1 + \beta) \left[\hat{u} \tanh\left(\frac{z_n \hat{u}}{\varpi}\right) + \bar{M} \tanh\left(\frac{z_n \bar{M}}{\varpi}\right) \right] \quad (56)$$

where $\varpi > 0$ and $\bar{M} > M / (1 - \beta)$. ϑ indicates the intermediate control variable. When the trigger rule (55) is satisfied, $u(t) = \vartheta(t_k)$ will be executed instantly and t_{k+1} will be computed. While in the interval $[t_k, t_{k+1})$, the actual controller u will remain at the value at t_k until the next trigger. From (54) and (55), we have $|\vartheta(t) - u(t)| < \beta |u(t)| + M$. Then, there exist two time-varying functions $\Psi_1, \Psi_2 \in [-1, 1]$ such that

$$u = \frac{\vartheta - M \Psi_2}{1 + \beta \Psi_1}. \quad (57)$$

Since $0 < 1 + \beta \Psi_1 \leq 1 + \beta$ and $-M \Psi_2 / (1 + \beta \Psi_1) \leq M / (1 - \beta)$, Combining (56) and (57) and Lemma 4, it follows that $z_n u \leq -|z_n \hat{u}| + 2K \varpi$.

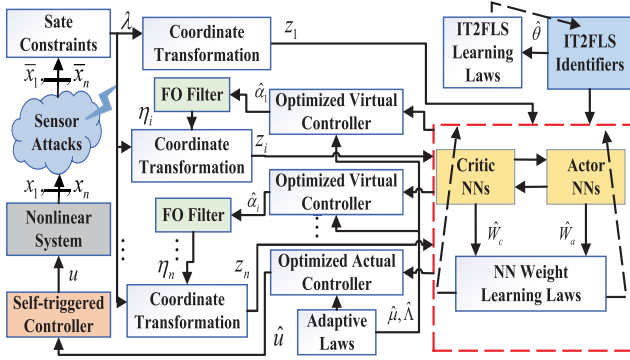


Fig. 1. Block diagram of the presented control approach.

In accordance with (8) and (9), there is

$$D^q \xi_n = -\hat{\Lambda}_n \tanh\left(\frac{\hat{\Lambda}_n \xi_n}{\varpi_n}\right) - r_n \xi_n + \bar{\Lambda}_n \quad (58)$$

where $|\bar{\Lambda}_n| = |-D^q \alpha_{n-1}| \leq \Lambda_n$ with $\tilde{\Lambda}_n = \hat{\Lambda}_n - \Lambda_n$.

Choose the Lyapunov function as $V_n = (1/2)z_n^2 + (1/2)\tilde{\mu}_n^2 + (1/2)\xi_n^2 + (1/2)\tilde{\Lambda}_n^2 + (1/2)\tilde{\theta}_n^T \tilde{\theta}_n + (1/2)\tilde{W}_{an}^T \tilde{W}_{an} + (1/2)\tilde{W}_{cn}^T \tilde{W}_{cn}$. Let $L_n = (1/2)z_n^2 + (1/2)\tilde{\mu}_n^2 + (1/2)\xi_n^2 + (1/2)\tilde{\Lambda}_n^2 + (1/2)\tilde{\theta}_n^T \tilde{\theta}_n$ and $\bar{L}_n = (1/2)\tilde{W}_{an}^T \tilde{W}_{an} + (1/2)\tilde{W}_{cn}^T \tilde{W}_{cn}$, one has

$$\begin{aligned} D^q L_n &\leq -k_n z_n^2 + z_n \zeta_{\lambda_n} \left(-\frac{1}{2} h \hat{W}_{an}^T \varphi_n + \varepsilon_n - \frac{\zeta_{\lambda_n}}{2} z_n \right) \\ &\quad - \frac{1}{4} h^2 \zeta_{\lambda_n}^2 z_n^2 - \hat{\Lambda}_n \xi_n \tanh\left(\frac{\hat{\Lambda}_n \xi_n}{\varpi_n}\right) - r_n \xi_n^2 \\ &\quad + |\xi_n| (\Lambda_n + \tilde{\Lambda}_n) - \sigma_n \tilde{\mu}_n \hat{\mu}_n - \ell_n \tilde{\Lambda}_n \hat{\Lambda}_n \\ &\quad - \rho_n \tilde{\theta}_n^T \hat{\theta}_n + 2h \zeta_{\lambda_n} K \varpi \end{aligned} \quad (59)$$

where $\tilde{\mu}_n = \hat{\mu}_n - \mu_n^*$ and $\tilde{\theta}_n = \hat{\theta}_n - \theta_n^*$.

Analogous to the previous analysis, one can get

$$\begin{aligned} D^q L_n &\leq -k_n z_n^2 + \frac{(\hat{W}_{an}^T \varphi_n)^2}{4} + \frac{\bar{\varepsilon}_n^2}{2} - r_n \xi_n^2 - \sigma_n \tilde{\mu}_n \hat{\mu}_n \\ &\quad - \ell_n \tilde{\Lambda}_n \hat{\Lambda}_n - \rho_n \tilde{\theta}_n^T \hat{\theta}_n + K \varpi_n + 2\bar{h} \delta K \varpi \end{aligned}$$

where $|\zeta_{\lambda_n}| \leq \delta$ and δ is a constant to be determined. Furthermore, a block diagram of the presented control approach is sketched in Fig. 1 to clarify the aforementioned design procedure.

B. Stability Analysis

Theorem 1: Under Assumptions 1, assuming that the optimal control scheme is adopted to the system (1) with unknown deceptive attacks and UFSC (5), including the optimal virtual controllers (22), (35) and (47), identifier-actor-critic weight learning laws (26)–(28), (37)–(39) and (49)–(51), adaptive laws (29), (40) and (41), and (52) and (53) along with the actual control laws (54) and (55), then the following attributes are ensured.

- 1) All the involved signals remain bounded in the CLS.

- 2) The stabilization error can converge to a small neighborhood of the origin at the minimum costs.
- 3) The UFSC without feasibility conditions are not violated.
- 4) The Zeno behavior can be precluded.

Proof 1: Taking (26) and (27), (37) and (38) and (49) and (50), $D^q \bar{L}_i$ ($1 \leq i \leq n$) yields

$$\begin{aligned} D^q \bar{L}_i &\leq -\gamma_{ai} \tilde{W}_{ai}^T \varphi_i^T \hat{W}_{ai} - \gamma_{ci} \tilde{W}_{ci}^T \varphi_i^T \hat{W}_{ci} \\ &\quad + (\gamma_{ai} - \gamma_{ci}) \tilde{W}_{ai}^T \varphi_i^T \hat{W}_{ci}. \end{aligned} \quad (60)$$

In view of $\hat{W}_{ai} - \tilde{W}_{ai} = \hat{W}_{ci} - \tilde{W}_{ci}$, one has

$$\tilde{W}_{ai}^T \varphi_i^T \hat{W}_{ai} = \frac{(\tilde{W}_{ai}^T \varphi_i)^2 + (\hat{W}_{ai}^T \varphi_i)^2 - (W_i^{*T} \varphi_i)^2}{2} \quad (61)$$

$$\tilde{W}_{ci}^T \varphi_i^T \hat{W}_{ci} = \frac{(\tilde{W}_{ci}^T \varphi_i)^2 + (\hat{W}_{ci}^T \varphi_i)^2 - (W_i^{*T} \varphi_i)^2}{2}. \quad (62)$$

Since the inequality $\tilde{W}_{ai}^T \varphi_i^T \hat{W}_{ci} \leq (\tilde{W}_{ai}^T \varphi_i)^2/2 + (\hat{W}_{ci}^T \varphi_i)^2/2$ holds, invoking (61) and (62) gets

$$\begin{aligned} D^q \bar{L}_i &\leq -\frac{\gamma_{ai}}{2} (\hat{W}_{ai}^T \varphi_i)^2 - \frac{\gamma_{ci}}{2} (\tilde{W}_{ci}^T \varphi_i)^2 - \frac{\gamma_{ci}}{2} (\tilde{W}_{ai}^T \varphi_i)^2 \\ &\quad - \left(\gamma_{ci} - \frac{\gamma_{ai}}{2} \right) (\hat{W}_{ci}^T \varphi_i)^2 + \frac{\gamma_{ai} + \gamma_{ci}}{2} (W_i^{*T} \varphi_i)^2. \end{aligned}$$

Then, select the global Lyapunov function to be $V = \sum_{i=1}^n V_i = \sum_{i=1}^n (L_i + \bar{L}_i)$, which produces

$$\begin{aligned} D^q V &\leq -\sum_{i=1}^n \left(k_i - \frac{1}{2} \right) z_i^2 - \sum_{i=1}^n \frac{\sigma_i}{2} \tilde{\mu}_i^2 - \sum_{i=2}^n r_i \xi_i^2 \\ &\quad - \sum_{i=2}^n \frac{\ell_i}{2} \tilde{\Lambda}_i^2 - \sum_{i=1}^n \frac{\rho_i}{2} \tilde{\theta}_i^2 - \sum_{i=1}^n \frac{\gamma_{ci}}{2} (\tilde{W}_{ai}^T \varphi_i)^2 \\ &\quad - \sum_{i=1}^n \frac{\gamma_{ci}}{2} (\tilde{W}_{ci}^T \varphi_i)^2 - \sum_{i=1}^n \frac{2\gamma_{ai} - 1}{4} (\hat{W}_{ai}^T \varphi_i)^2 \\ &\quad - \sum_{i=1}^n \frac{2\gamma_{ci} - \gamma_{ai}}{2} (\hat{W}_{ci}^T \varphi_i)^2 + \sum_{i=1}^n \frac{\sigma_i}{2} \mu_i^{*2} \\ &\quad + \sum_{i=1}^n \frac{\gamma_{ai} + \gamma_{ci}}{2} (W_i^{*T} \varphi_i)^2 + \sum_{i=2}^n \frac{\ell_i}{2} \Lambda_i^{*2} \\ &\quad + \sum_{i=1}^n \frac{\rho_i}{2} \theta_i^{*2} + \sum_{i=2}^n K \varpi_i + \sum_{i=1}^n \frac{\bar{\varepsilon}_i^2}{2} + 2\bar{h} \delta K \varpi. \end{aligned} \quad (63)$$

If choose $k_i > (1/2)$ and $\gamma_{ci} > (\gamma_{ai}/2) > (1/4)$, then

$$D^q V_n \leq -aV_n + b \quad (64)$$

where $a = \min\{2k_i - 1, 2r_i, \gamma_{ai} \lambda_{\min}^{\varphi_i}, \gamma_{ci} \lambda_{\min}^{\varphi_i}, \sigma_i, \ell_i, \rho_i\}$ and $b = \sum_{i=1}^n (\gamma_{ai} + \gamma_{ci}) (W_i^{*T} \varphi_i)^2/2 + 2\bar{h} \delta K \varpi + \sum_{i=1}^n \bar{\varepsilon}_i^2/2 + \sum_{i=2}^n K \varpi_i + \sum_{i=1}^n \sigma_i \mu_i^{*2}/2 + \sum_{i=2}^n \ell_i \Lambda_i^{*2}/2 + \sum_{i=1}^n \rho_i \theta_i^{*2}/2$. $\lambda_{\min}^{\varphi_i}$ signifies the minimum eigenvalue of matrix $\varphi_i \varphi_i^T$.

Based on Lemmas 1 and 3, for $\forall t \geq t_0$, one can get

$$V(t) \leq V(0) E_q(-at^q) + \frac{b}{a} \leq \frac{cV(0)}{1+at^q} + \frac{b}{a}.$$

For $t \geq ([cV(0)]/b)^{1/q}$, there is $V(t) \leq 2b/a$ as $t \rightarrow \infty$. We can further deduce that $|z_i| \leq (2(b/a)^{1/2})$, which implies

that the error variable is bounded. Similarly, the boundedness of $\tilde{\mu}_i, \xi_i, \tilde{\Lambda}_i, \tilde{\theta}_i, \tilde{W}_{ai}$ and $\tilde{W}_{ci}$ can be obtained. Given that $\mu_i^*, \Lambda_i, \theta_i^*, W_{ai}^*$ and W_{ci}^* are constants with the relationship $\tilde{\star} = \star - \star^*$, thus their estimated value $\hat{\mu}_i, \hat{\Lambda}_i, \hat{\theta}_i, \hat{W}_{ai}$ and $\hat{W}_{ci}$ are bounded. Accordingly, the boundedness of α_i, η_i and u can be derived. Then, from (5) and $\bar{x}_i(0) < c_i$ or $\bar{x}_i(0) > c_i$, one can get that $\bar{x}_i(t)$ and $x_i(t)$ are bounded as well. Thereby, one can obtain that all the signals in the CLS are bounded.

To proceed, as the boundedness of $u(t)$ can be confirmed, then the item $(\beta|u(t)| + M)/(\max\{l, |\dot{u}(t)|\})$ is proven to be bounded. Integrating (54) and (55), it is evident that there exists the interexecution interval ΔT with $\Delta T = t_{k+1} - t_k > 0$, implying that the Zeno behavior is successfully precluded. The proof is finished. ■

Remark 6: Theoretically, it can be derived from (64) that the stabilization error can be regulated as small as possible via increasing the relevant defined parameters in a or reducing those in b . However, a crucial fact is that over-tuning the parameters may increase control costs and even cause the adverse jitter phenomenon. Thereby, the appropriate parameters could be properly selected by designer to balance control effort and control performance through the trial-and-error strategy.

IV. SIMULATION STUDY

In this part, two explanatory analyses and several comparisons are offered to verify the efficacy and superiority of the reported optimal STC approach.

Example 1: Consider the fractional-order dynamic system described by

$$\begin{cases} D^{0.95} x_1 = x_2 + x_1 \sin(2x_1) \\ D^{0.95} x_2 = \cos^2(x_1) x_2 + u(t) \\ y = x_1. \end{cases} \quad (65)$$

A. Effectiveness Verification

The initial values are selected as $x_1(0) = 0, x_2(0) = -0.15, \hat{W}_{ai}(0) = \hat{W}_{ci}(0) = 0.1, \hat{\mu}_1(0) = \hat{\mu}_2(0) = \hat{\theta}_1(0) = \hat{\theta}_2(0) = \hat{\Lambda}_2(0) = 0$. The controller parameters are set as $\gamma_{ai} = 10, \gamma_{ci} = 8, c_1 = 0.5, c_2 = 0.3, k_1 = 8, k_2 = 10, \sigma_1 = 5, \sigma_2 = 10, \rho_1 = \rho_2 = \ell_2 = 3, M = 0.1, \bar{M} = 2, \beta = 0.3, \varpi = 0.1, l = 1.5, \varpi_i = 0.001, r_2 = 1$. It is assumed that the attack signal satisfy the relationship $\omega(t) = -0.3 - 0.3 \cos(t)$. Besides, the Gaussian functions is chosen as the radial basis function NN satisfying $\varphi_j(\tilde{z}_i) = \exp[-(\tilde{z}_i - d_j)^T(\tilde{z}_i - d_j)/1]$ with $i = 1, 2, j = 1, \dots, 9$ and $d_j = 0, \pm 1, \pm 2, \pm 3, \pm 4$. The fuzzy membership functions are given as $\underline{\mathfrak{s}}_{\tilde{p}_l^m} = 0.8 \exp[-((\tilde{s}_l - a_l^m)/4)^2/2]$, $\mathfrak{s}_{\tilde{p}_l^m} = \exp[-((\tilde{s}_l - a_l^m)/4)^2/2]$ with $l = 1, 2, m = 1, \dots, 5$ and $a_l^m = 0, \pm 1, \pm 2$.

The illustrative results of the put-forward methodology are presented in Figs. 2–4. More concretely, Fig. 2 displays the trajectories of the system state x_i and the compromised state $\bar{x}_i$ with $i = 1, 2$. From Fig. 2, it is observed that the system states can be effectively stabilized even though the constrained system is threatened by the unexpected FDIAs. Meanwhile, the curves of the norms of the actor-critic weights $\|\hat{W}_{ai}\|$ and $\|\hat{W}_{ci}\|$ are plotted in Fig. 3. Fig. 4 illustrates the trajectories of

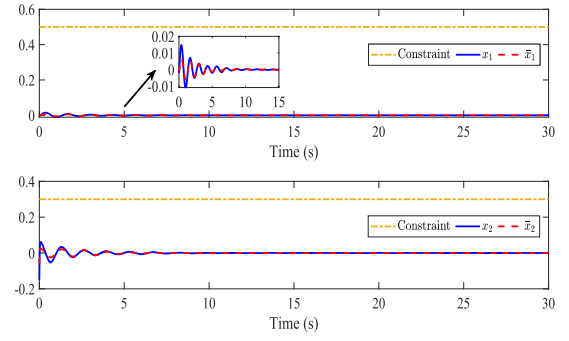


Fig. 2. Trajectories of x_i and $\bar{x}_i$.

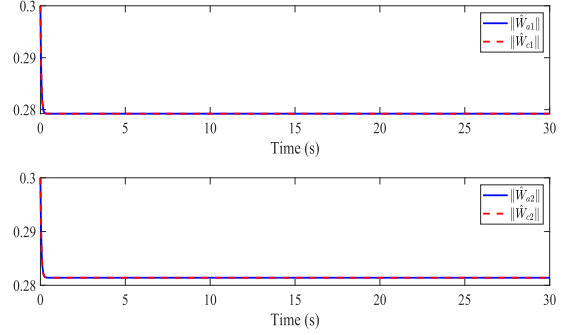


Fig. 3. Actor-critic NN weights.

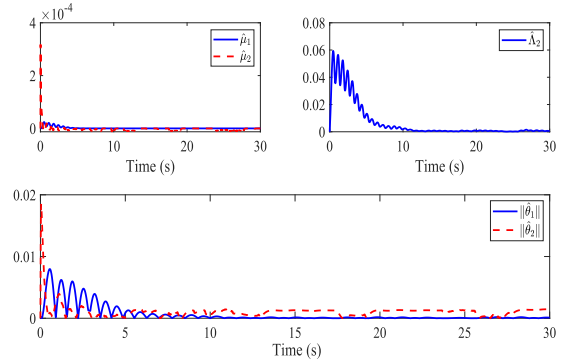


Fig. 4. Trajectories of adaptive parameters $\hat{\mu}_i, \hat{\Lambda}_2$, and $\|\hat{\theta}_i\|$.

adaptive parameters $\hat{\mu}_i, \hat{\Lambda}_2$ and $\|\hat{\theta}_i\|$. From Figs. 2–4, it can be drawn that all variables in the CLS are bounded.

B. Comparative Analysis

Additionally, some comparative results are given to demonstrate the superiority of the presented control program. First, the dynamic event-triggered mechanism (DETM) adopted in [26], as a comparison case with $\eta(0) = 0.4, c = d = \rho = 0.1$ and $\phi_1 = 2.2$, to show the difference between different trigger mechanisms. Figs. 5 and 6 display the curves of control input u and the trigger interval $\Delta T = t_{k+1} - t_k$ under different event-triggered mechanisms, which imply that less communication consumption can be achieved by the utilized STM. Besides, an optimal control solution studied in [18] and a BLF-based traditional adaptive backstepping control method discussed in [35] are employed to further clarify the merit of the put-forward control protocol. To ensure a valid comparison, the investigated system is assumed to be

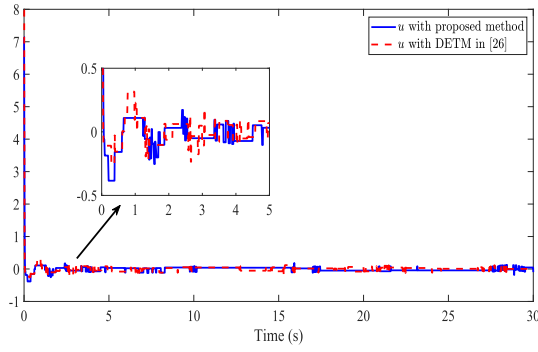
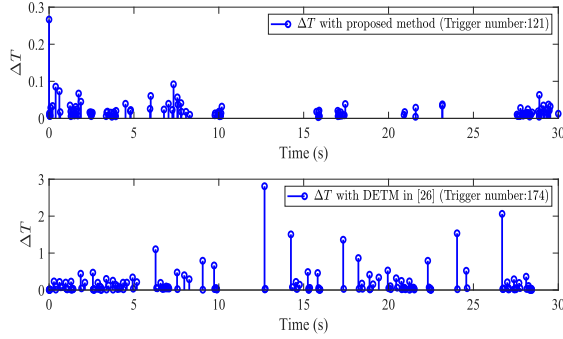
Fig. 5. Actual input u under different methods.Fig. 6. Trigger interval ΔT under different methods.

TABLE I

PERFORMANCE COMPARISON UNDER DIFFERENT METHODS

Performance indices	representation	method in [35]	Proposed
IAE	$\int_0^T z_1(t) dt$	0.02621	0.01847
ITAE	$\int_0^T t z_1(t) dt$	0.08131	0.06413
ISE	$\int_0^T z_1^2(t) dt$	0.00054	0.00044

free of FDIAs. In the first comparative group, all parameters involved in the proposed method are kept unchanged except for $x_2(0) = -0.1$, $c_2 = 0.5$, $k_2 = 5$, $r_2 = 15$ and $\bar{M} = 5$, while the related parameters in [18] are the same as the above selection. As depicted in Figs. 7 and 8, faster convergence for stabilization error and less performance cost can be ensured by the designed optimal controller with the UFSCs in comparison to the optimal controller devised in [18]. Then, in the second comparative set, the configuration of design parameters are $x_1(0) = -0.05$, $x_2(0) = 0.03$, $c_1 = k_{b1} = 0.8$, $c_2 = k_{b2} = 0.5$, $k_1 = 1$, $k_2 = 3$, $r_2 = 30$, $\rho_i = \delta_i = 3$, $a_1 = 0.5$, $a_2 = 1$, and others are the same as in Example 1. Actually, the constraint effect in the BLF-based control method relies on the variables of x_2 and α_1 due to the feasibility conditions $|z_2| < k_{b2}$, which leads to a deviation between the actual constraint performance and constraint objectives. Furthermore, three performance indicators including the integral absolute error (IAE), the integral time-weighted absolute error (ITAE), and the integral square error (ISE), are used to derive the quantifiable data as shown in Table I. According to Fig. 9 and Table I, it is acquired that faster convergence rate and higher control accuracy can be ensured by the presented algorithm under the equal constraints.

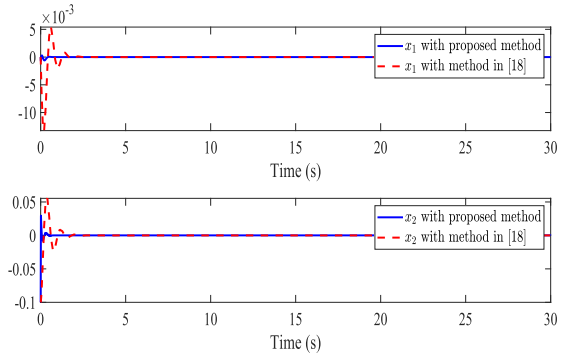
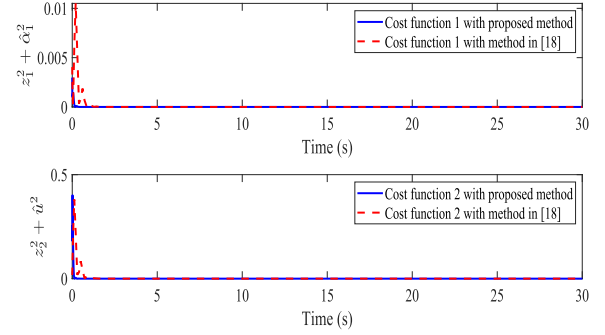
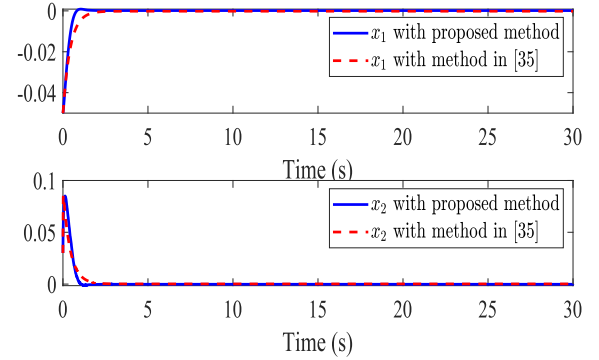
Fig. 7. Trajectories of x_i under different methods.

Fig. 8. Cost functions under different methods.

Fig. 9. Trajectories of x_i under different methods.

Example 2: In this example, a single-machine-infinite bus (SMIB) power system depicted in Fig. 10 is employed to further confirm the efficacy and practicability of the proposed control solution. According to [41] and [48], the mathematical expression of fractional-order SMIB power system is

$$\begin{cases} D^q \check{\varphi} = \check{\omega} \\ D^q \check{\omega} = -\frac{\mathcal{F}}{\mathcal{L}} \check{\omega} - \frac{\mathcal{P}_{\max}}{\mathcal{L}} \sin(\check{\varphi}) + \frac{\mathcal{P}_m}{\mathcal{L}} + \frac{\mathcal{P}_a}{\mathcal{L}} \cos(\epsilon t) + u(t) \end{cases} \quad (66)$$

where q is the fractional order, $\check{\varphi}$ and $\check{\omega}$ mean the relative angle and speed between two generators, respectively; $\mathcal{F}$ denotes the damping coefficient and $\mathcal{L}$ represents the rotational inertia of generators; $\mathcal{P}_{\max}$ stands for the generator's maximum power; $\mathcal{P}_m$ and $\mathcal{P}_a$ indicate the power of the machine and the power perturbation's amplitude. Let $x_1 = \check{\varphi}$ and $x_2 = \check{\omega}$, (66) can be

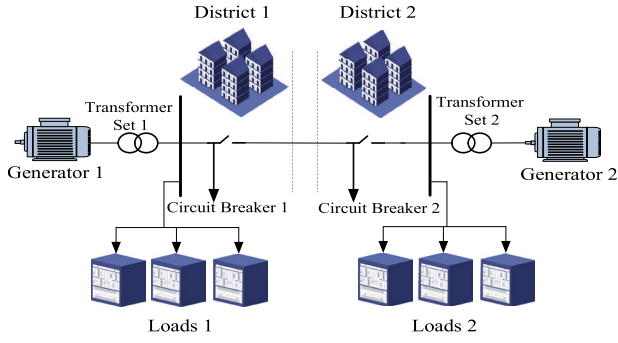


Fig. 10. Structure of SMIB power system.

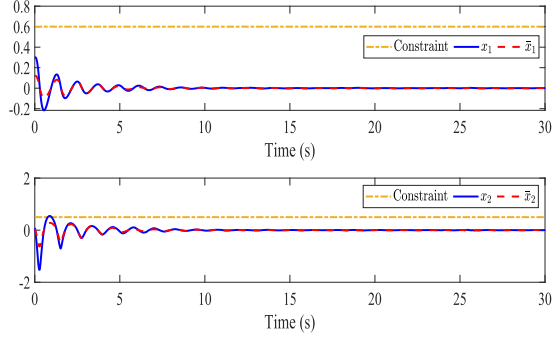
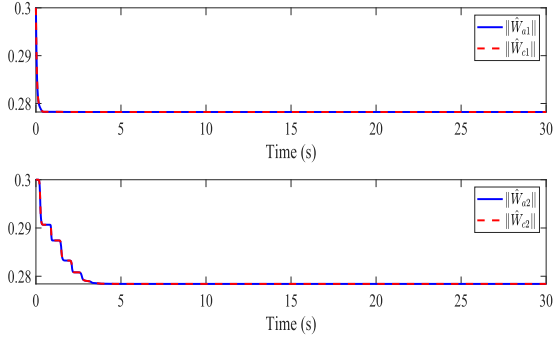
Fig. 11. Trajectories of x_i and $\bar{x}_i$.

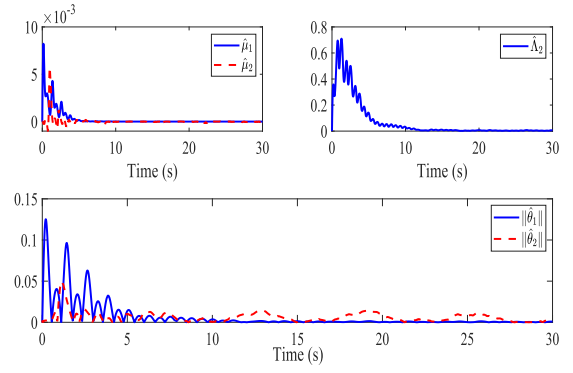
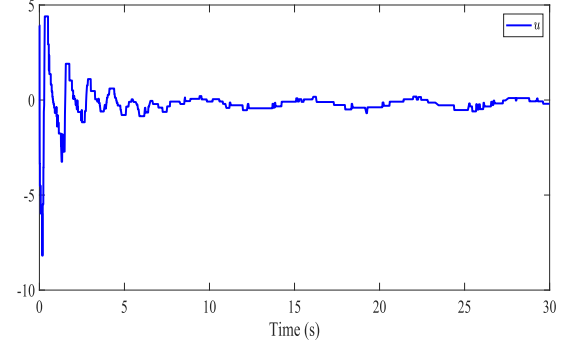
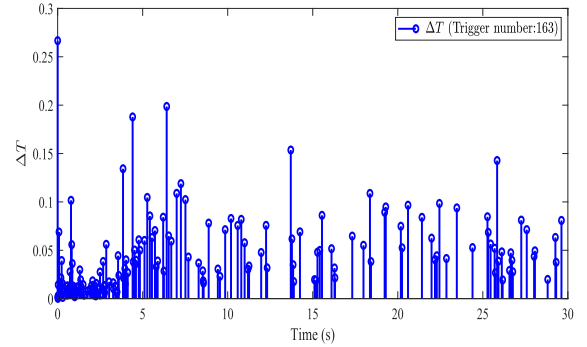
Fig. 12. Actor-critic NN weights.

rewritten as

$$\begin{cases} D^q x_1 = x_2 \\ D^q x_2 = -0.02x_2 - \sin(x_1) + 0.2 \\ \quad + 0.2593 \cos(t) + u(t) \end{cases} \quad (67)$$

where $q = 0.95$, $\mathcal{F}/\mathcal{L} = 0.02$, $\mathcal{P}_{\max}/\mathcal{L} = 1$, $\mathcal{P}_m/\mathcal{L} = 0.2$, $\mathcal{P}_a/\mathcal{L} = 0.2593$, $\epsilon = 1$.

The initial condition is set as $x_1(0) = 0.3$, $x_2(0) = 0$, the design parameters are $c_1 = 0.6$, $c_2 = 0.5$, $k_1 = 10$, $k_2 = 15$, and $\bar{M} = 1.5$, and the other scales are selected as same as that in Example 1. The simulation results are presented in Figs. 11–15. Among them, Fig. 11 depicts the curves of the system state x_i ($i = 1, 2$) and the compromised state $\bar{x}_i$ influenced by the attack signals, respectively. It is easily obtained from Fig. 11 that a relatively satisfactory stabilization result can be guaranteed even if the FDIAs exert a degree of adverse impacts on control performance, and the state

Fig. 13. Trajectories of adaptive parameters $\hat{\mu}_i$, $\hat{\lambda}_2$ and $\|\hat{\theta}_i\|$.Fig. 14. Actual input u .Fig. 15. Trigger interval $\Delta T = t_{k+1} - t_k$.

variables $\bar{x}_i$ strictly adhere to the pregiven constraints $c_1 = 0.6$ and $c_2 = 0.5$. Further, the time responses of actor-critic weights are displayed in Fig. 12. Fig. 13 reveals the curves of the adaptive parameters $\hat{\mu}_i$, $\hat{\lambda}_2$ and the norm of $\hat{\theta}_i$. In Figs. 14 and 15, the dynamic behavior of the control signal u and the interevent interval ΔT are exhibited. In addition, it is also verified that the boundedness of all the signals in the CLS can be ensured by using the proposed controller through the above simulation results.

V. CONCLUSION

An RL-based adaptive fuzzy optimized secure STC algorithm was presented for constrained FONSs under unknown FDIAs. Initially, a nonlinear state-dependent function with convexity is applied to fulfill the UFSCs and abolish the additional feasibility conditions. Moreover, by incorporating the equivalent auxiliary systems and STM into optimized

recursive control design, an adaptive optimal secure control paradigm is established to achieve fractional optimal control, which can evade the adverse impacts of FDIAs on feedback design and diminish the communication consumption. Additionally, it has been proved that all the internal signals in the CLS are bounded and the stabilization errors can be regulated into a compact set nearby the origin by means of the fractional Lyapunov stability theory. Eventually, simulation results and comparative analyses are developed to substantiate the validity and practicality of the introduced control solution. Admittedly, the proposed control tactic is confined to local situation that all state information are assumed to be available and the specific constraints are not taken into account. In the future, we will devote to developing an adaptive optimal safety-critical control algorithms for constrained FONSs with unmeasurable states and malicious cyberattacks.

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